

Hotel Operations Analysis – Solution Guide

Basic Questions

Q1. What is the average lead time for bookings?

Answer: The average lead time for hotel bookings is **79.94 days**, indicating that guests typically make reservations about 80 days before their arrival date.

Q2. What is the distribution of bookings by hotel type?

Answer: The dataset contains:

- **City Hotel:** 53,269 bookings
- **Resort Hotel:** 33,938 bookings

City Hotels account for the majority of reservations.

Q3. How many bookings were canceled?

Answer: A total of **23,996 bookings** were canceled.

Q4. What is the most common arrival month for bookings?

Answer: **August** is the most common arrival month with **11,242 bookings**, making it the busiest month for hotel reservations.

Q5. What is the average number of special requests per booking?

Answer: Guests made an average of **0.70 special requests per booking**.

Q6. Which country has the highest number of bookings?

Answer: **Portugal** generated the highest number of bookings with **27,337 reservations**.

Q7. What is the average daily rate (ADR) for each hotel type?

Answer:

- **City Hotel:** 111.17
- **Resort Hotel:** 99.10

City Hotels have a higher average room rate than Resort Hotels.

Q8. What percentage of guests required car parking spaces?

Answer: 7,306 bookings required parking spaces, representing 8.38% of total bookings.

Q9. What is the average stay duration in week nights and weekend nights?

Answer:

- Average Week Night Stay: **2.62 nights**
- Average Weekend Night Stay: **1.00 night**

Guests typically stay longer during weekdays than weekends.

Q10. How many bookings were made through travel agents?

Answer: 75,077 bookings were made through travel agents, accounting for 86.09% of all reservations.

Intermediate Questions

Q1. What is the cancellation rate for each hotel type?

Answer:

- **City Hotel:** 30.10%
- **Resort Hotel:** 23.46%

City Hotels experience a higher cancellation rate than Resort Hotels.

Q2. What is the average ADR per market segment?

Answer:

Market Segment ADR

Online TA	118.29
Direct	116.85
Aviation	100.61
Offline TA/TO	81.59
Groups	75.18
Corporate	68.37
Undefined	15.00
Complementary	3.09

Online TA generates the highest average ADR.

Q3. What is the relationship between lead time and cancellation rate?

Answer: The correlation between lead time and cancellation is **0.184**, indicating a weak positive relationship. Guests who book further in advance are slightly more likely to cancel.

Q4. Which distribution channel has the highest number of bookings?

Answer:

Distribution Channel Bookings

TA/TO	69,018
Direct	12,946
Corporate	5,057
GDS	181
Undefined	5

TA/TO is the dominant distribution channel.

Q5. What is the average number of previous cancellations by hotel type?

Answer:

- **City Hotel:** 0.04
- **Resort Hotel:** 0.02

City Hotel guests have a slightly higher history of previous cancellations.

Q6. What is the trend of ADR over the years?

Answer:

Year ADR

2015	92.45
2016	101.57
2017	118.92

ADR increased consistently over the years.

Q7. Which month has the highest revenue?

Answer: August generated the highest revenue with **1,698,163.63**.

Q8. What is the impact of special requests on ADR?

Answer: ADR increases as the number of special requests increases.

Special Requests ADR

0	99.80
1	109.78
2	118.70
3	125.23
4	131.09
5	129.98

Correlation between special requests and ADR = **0.146**.

Q9. What is the average stay duration for repeated guests versus new guests?

Answer:

- **New Guests:** 3.70 nights
- **Repeated Guests:** 1.95 nights

New guests stay longer than repeated guests.

Q10. Which room type has the highest number of bookings?

Answer: Room Type A has **56,417 bookings**, representing **64.69%** of total reservations.

Advanced Questions

Q1. What factors significantly impact the cancellation rate?

Answer: Logistic Regression analysis identified:

Feature	Coefficient
Previous Cancellations	0.605
ADR	0.007
Lead Time	0.005
Total Special Requests	-0.461
Booking Changes	-0.497

Model Accuracy: **73.79%**

Previous cancellations are the strongest predictor of future cancellations.

Q2. How does the ADR vary with the number of adults, children, and babies?

Answer:

Feature	Coefficient
Children	38.39
Adults	32.86
Babies	7.61

R² Score: **0.214**

Children have the strongest impact on ADR.

Q3. What is the impact of booking changes on guest satisfaction as indicated by special requests?

Answer:

- Correlation = **0.018**
- Regression Coefficient = **0.021**
- R² Score = **0.0003**

There is virtually no meaningful relationship between booking changes and special requests.

Q4. What is the seasonal impact on booking cancellations?

Answer:

Month	Cancellation Rate (%)
January	22.14
February	23.22
March	24.42
April	30.46
May	29.27
June	30.34
July	31.83
August	32.22
September	24.47
October	23.68
November	21.16
December	26.94

Highest: **August (32.22%)**

Lowest: **November (21.16%)**

Q5. How does the booking lead time distribution vary between different market segments?

Answer:

Market Segment	Average Lead Time (Days)
Groups	147.91
Offline TA/TO	106.08
Online TA	79.97
Direct	48.69
Corporate	16.27
Complementary	13.82
Aviation	4.30
Undefined	1.50

The **Groups** segment has the highest average lead time, indicating that group reservations are typically planned far in advance.